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$$\int_{\pi}^{2\pi} \cos^2(x) dx$$

Eerst de integraal $\int \cos^2(x) dx$ oplossen:

$$\begin{aligned} & \int \cos^2(x) dx \\ &= \int \frac{1 + \cos(2x)}{2} dx \\ &= \int \frac{1}{2} dx + \frac{1}{2} \int \cos(2x) dx \end{aligned}$$

$$t = 2x$$

$$dt = 2dx$$

$$\begin{aligned} &= \frac{x}{2} + \frac{1}{4} \int \cos(t) dx \\ &= \frac{x}{2} + \frac{\sin(t)}{4} + C \\ &= \frac{x}{2} + \frac{\sin(2x)}{4} + C \end{aligned}$$

Nu met de grenzen!:

$$\begin{aligned} &= \frac{2\pi}{2} + \frac{\sin(4\pi)}{4} - \frac{\pi}{2} - \frac{\sin(2\pi)}{4} \\ &= \pi + 0 - \frac{\pi}{2} - 0 = \frac{\pi}{2} \end{aligned}$$

References